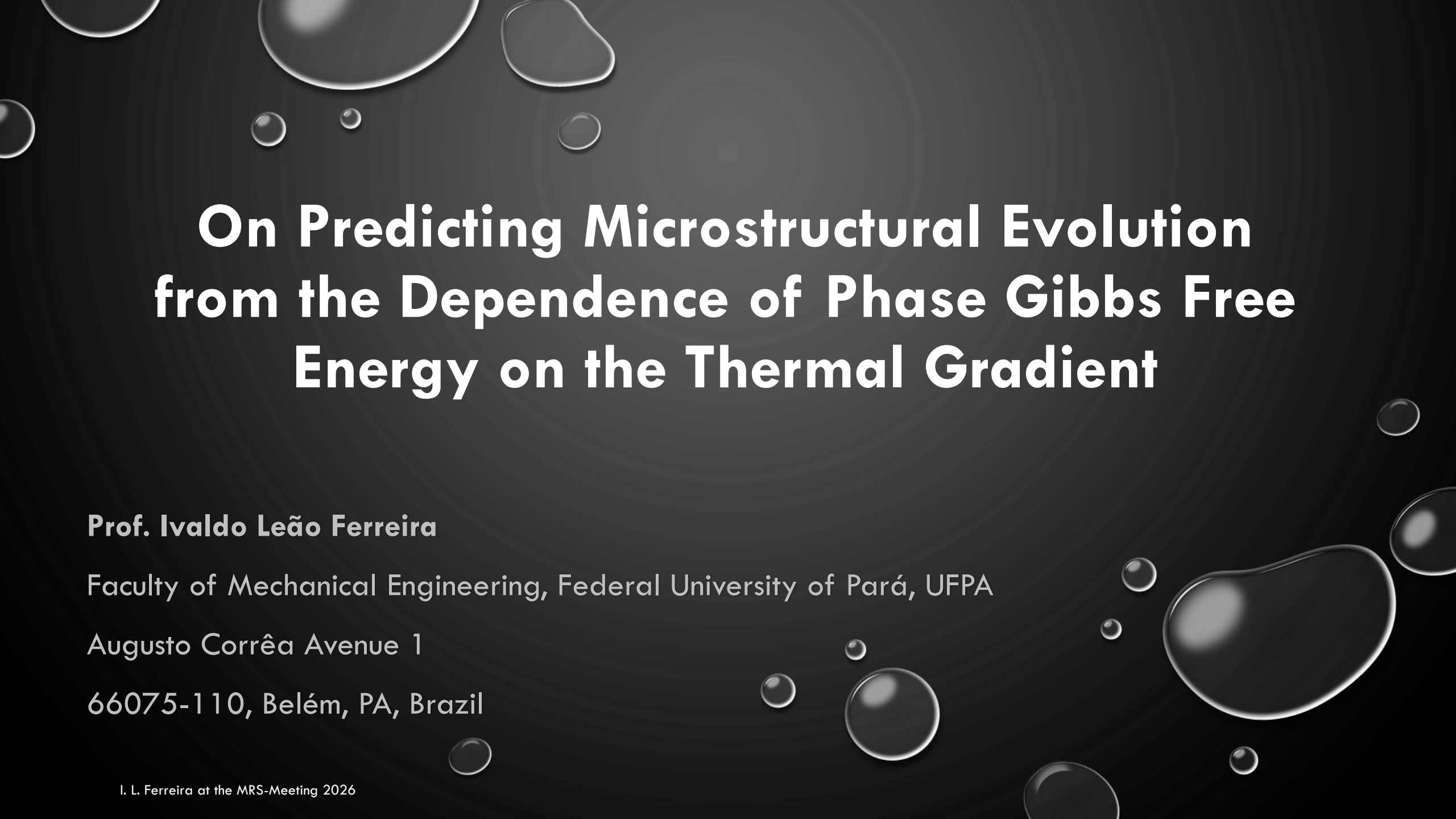


The background of the slide is dark gray with several realistic water droplets of various sizes scattered across it. Some droplets are at the top, some on the right, and some near the bottom. The main title is centered in a large, bold, white font.

On Predicting Microstructural Evolution from the Dependence of Phase Gibbs Free Energy on the Thermal Gradient

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MOTIVATION

- Our experimental **cooling rates** range from **2100 to 120 K/min**, whereas **equilibrium** is reached at **less than 5 K/min**;
- The **Classical Nucleation Theory (CNT)** provides an inadequate description of thermodynamic equilibrium shifts;
- The CNT's core equation – a limiting case for infinite volume – **overemphasizes bulk free energy** contributions when considering finite surface energies, i.e., $\lim_{r \rightarrow +\infty} \frac{\Delta G_S}{\Delta G_V} = 3/r$;
- The **undercooling** is **dependent on the thermal gradient** and surface free-energy changes considerably;
- Recent research by Ferreira (2024) and Ferreira et al. (2025) on phase nucleation demonstrates a closed-form **parabolic relationship** in terms of **the nucleation radius**.

OBJECTIVES

- Remain in the continuum mechanics scale to easily calculate the surface stress tensor and make DTA/DSC experiments for equilibrium conditions;
- Derive a general phase transformation equation within this domain by combining the **First** and **Second** Laws of Thermodynamics;
- Present a novel formulation of phase nucleation;
- Apply fundamental thermodynamic variables and functions to analyze this new field;
- **Investigate the effects of thermal gradients on free Energies;**

A Proper Field for Phase Transformation

- The First Law of Thermodynamics,

$$\delta Q - \delta W = \Delta E$$

- This means, $Q = f(E, W)$. Energy E is a **point function** and, Q and W are **path-dependent quantities**, i.e., line functions

$$\delta Q - \delta W = dE$$

- **Let's then define a new field**, by making from tensor calculus, $dE \equiv \partial E, i$, in relation to T primitive variable, we can express the **1st Law of Thermodynamics** as

$$\nabla \left[\frac{\partial T}{\partial Q} (Q - W) \right] = \nabla \left[\frac{\partial T}{\partial Q} E \right]$$

A Proper Field for Phase Transformation

The thermal field T can be defined as:

- Path-independent thermal field gradient

$$\nabla T_{\text{pi}} = \nabla \left[\frac{\partial T}{\partial Q} (Q - W) \right] = \nabla \left[\frac{\partial T}{\partial Q} E \right]$$

- Path-dependent thermal field gradient

$$\nabla T_{\text{pd}} = \nabla \left[\frac{\partial T}{\partial Q} Q \right] = \nabla \left[\frac{\partial T}{\partial Q} E \right] + \nabla \left[\frac{\partial T}{\partial Q} W \right]$$

- Thermal-fields relationship gradient

$$\nabla T_{\text{pd}} = \nabla \left[\frac{\partial T}{\partial Q} Q \right] = \nabla T_{\text{pi}} + \nabla \left[\frac{\partial T}{\partial Q} W \right]$$

A Proper Field for Phase Transformation

By considering LHS and RHS, the following expression can be deduced,

$$\nabla \left(\frac{\partial T}{\partial Q} Q \right) = \nabla \left[\frac{\partial T}{\partial Q} (E + W) \right] \cong \nabla \left[\frac{\partial T}{\partial E + \delta W} E \right] + \nabla \left[\frac{\partial T}{\delta E + \delta W} W \right]$$

By the First Law, $\delta Q = f(dE, \delta W)$, presents E and W as **independent variables**, $\nabla \left[\frac{\partial T}{\delta W} E \right]$ and $\nabla \left[\frac{\partial T}{\partial E} W \right]$ **can be neglected**. Finally, the **path-dependent thermal field $\mathbf{T}$** in [K] can be defined as,

$$\nabla \mathbf{T}_{pd} = \nabla \left(\frac{\partial T}{\partial Q} Q \right)$$

and, field derived in 2024 in Ferreira [2024] DOI: [10.1016/j.physb.2024.416494](https://doi.org/10.1016/j.physb.2024.416494)

$$\nabla \mathbf{T} = \nabla \left[\frac{\partial T}{\partial Q} (E + W) \right] \cong \nabla \left[\frac{\partial T}{\partial E} E \right] + \nabla \left[\frac{\partial T}{\delta W} W \right]$$

A Proper Field for Phase Transformation

The energy can be written in terms of $E = U + K + E_P + \Sigma_{other}$, (EMMY NOETHER'S SIMMETRY)

$$\nabla \mathbf{T} \cong \nabla \left[\frac{\partial T}{\partial E} E \right] + \nabla \left[\frac{\partial T}{\partial W} W \right] \cong \nabla \left[\frac{\partial T}{\partial E} (U + K + E_P + \Sigma_{other}) \right] + \nabla \left[\frac{\partial T}{\partial W} W \right]$$

By the First Law,

$$\delta Q - \delta W = dE = dU + dK + dE_P + d\Sigma_{other}$$

where, U can be written as $U = f(T, V, P, n_1 \dots n_N)$ to give $dU = \frac{\partial U}{\partial T} dT + \frac{\partial U}{\partial V} dV + \frac{\partial U}{\partial P} dP + \sum_i^N \frac{\partial U}{\partial n_i} dn_i$,

$$\nabla \mathbf{T} \cong \nabla \left[\frac{\partial T}{\partial E} E \right] + \nabla \left[\frac{\partial T}{\partial W} W \right] \cong \nabla \left[\frac{\partial T}{(\partial U + \partial K + \partial E_P + \partial \Sigma_{other})} (U + K + E_P + \Sigma_{other}) \right] + \nabla \left[\frac{\partial T}{\partial W} W \right]$$

A Novel Phase Nucleation Formulation

- The critical total free energy ΔG in Ferreira [2024] DOI: 10.1016/j.physb.2024.416494 can be deduced in terms of the bulk ΔG_V and surface ΔG_S free energies as,

$$\Delta G = \left(\frac{1}{3} \pi r_C^3 \Delta S_V \Delta T + \pi r_C^2 \gamma_{SL} \right) f(\theta) = \left(\frac{1}{3} \pi r_C^3 \Delta G_V + \pi r_C^2 \Delta G_S \right) f(\theta)$$

where, $f(\theta) = 2 - 3 \cos \theta + \cos^3 \theta$. When deriving with respect to the critical radius r_C to determine the maximum r_C of ΔG_C , the following parabolic equation is obtained,

$$\left[\left(\frac{\partial \Delta S_V}{\partial r} \Delta T + \Delta S_V \frac{\partial \Delta T}{\partial r} \right) f(\theta) + \Delta S_V \Delta T \frac{\partial f(\theta)}{\partial r} \right] r_C^2 + 3 \left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) f(\theta) + \gamma_{SL} \frac{\partial f(\theta)}{\partial r} \right] r_C + 6 \gamma_{SL} f(\theta) = 0$$

A Novel Phase Nucleation Formulation

- A **2nd-order critical radius** r_C^{2nd} can be derived from nucleation formulation in terms of bulk ΔG_V , surface ΔG_S , and configurational ΔG_C free energies as given,

$$r_C^{2nd} = -\frac{3}{2} \frac{\left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) + \frac{\gamma_{SL}}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]}{\left[\left(\frac{\partial \Delta S_V}{\partial r} \Delta T + \Delta S_V \frac{\partial \Delta T}{\partial r} \right) + \frac{\Delta S_V \Delta T}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]} = -\frac{3}{2} \frac{\left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) f(\theta) + \gamma_{SL} \frac{\partial f(\theta)}{\partial r} \right]}{\left[\frac{\partial (\Delta S_V \Delta T f(\theta))}{\partial r} \right]} = -\frac{3}{2} \frac{\left[\Delta G_V + \frac{\partial \Delta G_S}{\partial r} + \Delta G_C \right]}{\frac{\partial \Delta G_V}{\partial r}}$$

and, the **1st-order approximation of critical radius** r_C^{1st} can be expressed as the following

$$r_C^{1st} = -\frac{2\gamma_{SL}f(\theta)}{\left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) f(\theta) + \gamma_{SL} \frac{\partial f(\theta)}{\partial r} \right]} = -\frac{2\Delta G_S}{\Delta G_V + \frac{\partial \Delta G_S}{\partial r} + \Delta G_C}$$

By comparing with CNT, $r_C^{CNT} = -\frac{2\Delta G_S}{\Delta G_V}$

A Novel Phase Nucleation Formulation

- A 2nd-order thermal field tensor Γ^{2nd} can be derived by maximum of total Gibbs-free energy,

$$\Gamma^{2nd} = -\frac{3}{4} \frac{\left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) + \frac{\gamma_{SL}}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right] \Delta T}{\left[\left(\frac{\partial \Delta S_V}{\partial r} \Delta T + \Delta S_V \frac{\partial \Delta T}{\partial r} \right) + \frac{\Delta S_V \Delta T}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]} = -\frac{3}{4} \frac{\left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) f(\theta) + \gamma_{SL} \frac{\partial f(\theta)}{\partial r} \right]}{\frac{1}{\Delta T} \left[\frac{\partial (\Delta S_V \Delta T f(\theta))}{\partial r} \right]} = -\frac{3}{4} \frac{\left[\Delta G_V + \frac{\partial \Delta G_S}{\partial r} + \Delta G_C \right]}{\frac{\partial \Delta S_V}{\partial r}}$$

And 1st-order approximation for the thermal field tensor Γ^{1st} can be derived as given,

$$\Gamma^{1st} = -\frac{\gamma_{SL} f(\theta)}{\left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) f(\theta) + \gamma_{SL} \frac{\partial f(\theta)}{\partial r} \right]} \Delta T = -\frac{\Delta G_S}{\Delta G_V + \frac{\partial \Delta G_S}{\partial r} + \Delta G_C} \Delta T = -\frac{\Delta G_S}{\frac{1}{\Delta T} \left[\Delta G_V + \frac{\partial \Delta G_S}{\partial r} + \Delta G_C \right]} = -\frac{\Delta G_S}{\Delta S_V + \frac{\partial \Delta S_S}{\partial r} + \Delta S_C}$$

- The nucleation rate I derived by Ferreira (2024) is given by

$$I = \frac{DA}{\lambda^4} \frac{N}{V} \exp \left(\frac{\Delta G_C}{\Delta G_{C,Eq}} \right)$$

where, D is the self-diffusion coefficient in m^2/s , N/V is the density of modes in the new phase in $modes/m^3$, λ is the distance an atom must jump from a phase to the nucleating new surface area in m, A is the surface area of the new phase in m^2 .

A General Phase Transformation Equation

- The **newly thermal field Tensor Γ** as derived by Ferreira [2024] DOI: 10.1016/j.physb.2024.416494 is defined as any new created and deformed surface area $\mathbf{A}$ is normal to the thermal field $\mathbf{T}$, i.e.,

$$\Gamma = \mathbf{A} \cdot \nabla \mathbf{T} = \mathbf{A} \cdot \nabla \mathbf{T}_{pi} = \mathbf{A} \cdot \nabla \left[\frac{\partial T}{\partial Q} (Q - W) \right] \cong \mathbf{A} \cdot \nabla \left[\frac{\partial T}{\partial E} E \right]$$

- Examples of application,**

$$\Gamma = -\frac{3}{4} \frac{\left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) + \frac{\gamma_{SL}}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right] \Delta T}{\left[\left(\frac{\partial \Delta S_V}{\partial r} \Delta T + \Delta S_V \frac{\partial \Delta T}{\partial r} \right) + \frac{\Delta S_V \Delta T}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]} = -\frac{3}{4} \frac{\left[\Delta G_V + \frac{\partial \Delta G_S}{\partial r} + \Delta G_C \right]}{\frac{\partial \Delta S_V}{\partial r}} = \mathbf{A} \cdot \nabla \mathbf{T}$$

$$\nabla \mathbf{T} \cong \nabla \left[\frac{\partial T}{\partial U} (U_{transl} + \mathbf{U}_{vib} + U_{rot} + \mathbf{U}_{elect} + U_{mag} + U_{other}) \right] + \nabla \left[\frac{\partial T}{\partial K} K \right] + \nabla \left[\frac{\partial T}{\partial E_P} E_P \right] + \nabla \left[\frac{\partial T}{\partial \Sigma_{other}} \Sigma_{other} \right] + \nabla \left[\frac{\partial T}{\partial W} W \right]$$

Application of Fundamental Thermodynamics Variables and Functions to this New Field

The solution of the thermal field tensor Γ requires:

- **The solution of the surface-stress tensor Σ** [Gurtin and Murdoch, 1978; Ferreira (2022)]

$$\nabla_S \cdot \Sigma = \Theta \mathbf{n} + \rho_0 \ddot{\mathbf{u}}$$

$$\Sigma = \sigma \mathbf{I} + 2(\mu_0 - \sigma) \mathbf{E} + (\lambda_0 - \sigma)(\text{tr } \mathbf{E}) \mathbf{I} + \sigma \nabla_S \mathbf{u} \text{ on surface } S$$

$$\mathbf{E} = \frac{1}{2}(D\mathbf{u} + D\mathbf{u}^T)$$

where Θ , $\mathbf{E}$ and $\mathbf{u}$ are the stress, the strain and displacement fields.

- **A metric tensor, in this case, in spherical coordinates**, and its contravariant,

$$g_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}, \text{ and the initial conditions, } g_0^{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r_0^{-2} & 0 \\ 0 & 0 & r_0^{-2} \sin^{-2} \theta_0 \end{pmatrix}$$

Application of Fundamental Thermodynamics Variables and Functions to this New Field

The solution of the thermal field Tensor Γ requires:

- **The Surface Free Energy** [Ferreira, (2021, 2022, and 2024)]:

$$\Delta G_S = \gamma_0 \Phi(r_0^2 R) - \frac{1}{2A} \int_s \Sigma^{ij} g_{ij} dA$$

Considering a 2-Sphere, the Ricci scalar is $R = 2/r_0^2$,

$$\Delta G_S^{sph} = \Delta G_{S0} \frac{3}{1 + \frac{r^2}{r_0^2} + \frac{r^2 \sin^2 \theta}{r_0^2 \sin^2 \theta_0}} - \frac{1}{4\pi r_0^2} \int_{A_0}^{A_1} \Sigma dA$$

where Φ is a dimensionless geometric function; its argument $r_0^2 R$ is the only admissible scalar curvature invariant, $\Delta G_{S0} = \gamma_0$, r_0 , Σ , and R are the surface energy in [$J.m^{-2}$], the radius at the thermodynamic state 0 (assumed as equilibrium or initial condition vector radius for Equilibrium, i.e., $\partial\Omega = \dots_{ij}$) in [m], the surface stress tensor in [$N.m^{-1}$], and the Ricci scalar, respectively.

Application of Fundamental Thermodynamics Variables and Functions to this New Field

The solution of the thermal field Tensor Γ requires:

- **The Surface Entropy** [Ferreira, (2021, 2022, and 2024)]:

$$\Delta S_S = \frac{\gamma_0}{\Delta T} \Phi(r_0^2 R) - \frac{1}{2A\Delta T} \int_s \Sigma^{ij} g_{ij} dA$$

Considering a 2-Sphere, the Ricci scalar is $R = 2/r_0^2$,

$$\Delta S_S^{sph} = \frac{\Delta G_{S0}}{\Delta T} \frac{3}{1 + \frac{r^2}{r_0^2} + \frac{r^2 \sin^2 \theta}{r_0^2 \sin^2 \theta_0}} - \frac{1}{\Delta T 4\pi r_0^2} \int_{A_0}^{A_1} \Sigma dA$$

where $\Delta G_{S0} = \gamma_0$, r_0 , Σ , and R are the surface energy in $[J.m^{-2}]$, the radius at the thermodynamic state 0 (assumed as equilibrium or initial condition vector radius for Equilibrium, i.e., $\partial\Omega = g_0 \quad ij$) in $[m]$, the surface stress tensor in $[N.m^{-1}]$, and the Ricci scalar, respectively.

Application of Fundamental Thermodynamics Variables and Functions to this New Field

The solution of the thermal field Tensor Γ requires:

- **The Surface Energy Potential** [Ferreira, (2021, 2022, and 2024)]:

$$\frac{\partial \Delta G_S}{\partial r} = \partial_k \left[\gamma_0 \Phi(r_0^2 R) - \frac{1}{2A} \int_s \Sigma^{ij} g_{ij} dA \right] = \Delta G_{S0} (1 + r_0^2 R) \frac{g_0^{ij} \nabla_k g_{ij}}{(g_{mn} g_0^{mn})^2} - \frac{1}{2A} \partial_k \int_s \Sigma^{ij} g_{ij} dA$$

Considering a 2-Sphere, the Ricci scalar is $R = 2/r_0^2$, since $\nabla_k R = 0$, $\nabla_k g_0^{ij} = 0$, and $k = r$

$$\left. \frac{\partial \Delta G_S}{\partial r} \right|^{sph} = \Delta G_{S0} \frac{-6 \left[\frac{r \sin^2 \theta}{r_0^2 \sin^2 \theta_0} + \frac{r}{r_0^2} \right]}{\left(1 + \frac{r^2}{r_0^2} + \frac{r^2 \sin^2 \theta}{r_0^2 \sin^2 \theta_0} \right)^2} - \frac{1}{\frac{r_0^2}{r^2}} \Sigma_r(r)$$

where $\Delta G_{S0} = \gamma_0$, r_0 , Σ , and R are the surface energy in [$J.m^{-2}$], the radius at the thermodynamic state 0 (assumed as equilibrium or initial condition vector radius for Equilibrium, i.e., $\partial \Omega = g_0 \quad_{ij}$) in [m], the surface stress tensor in [$N.m^{-1}$], and the Ricci scalar, respectively.

Application of Fundamental Thermodynamics Variables and Functions to this New Field

The solution of the thermal field Tensor Γ requires:

- **The Surface Entropy Potential** [Ferreira, (2021, 2022, and 2024)]:

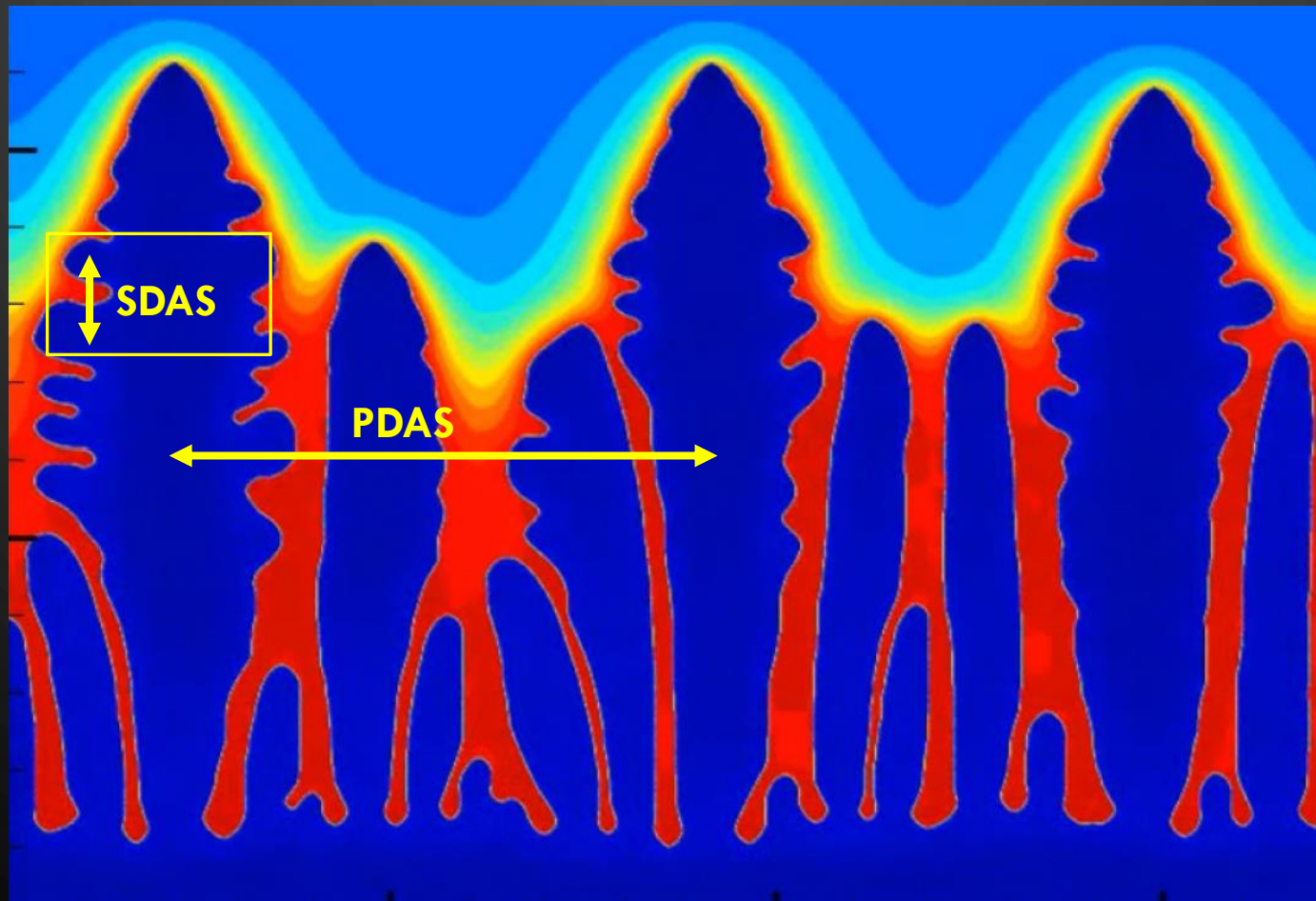
$$\frac{\partial \Delta S_S}{\partial r} = \frac{1}{\Delta T} \partial_k \left[\gamma_0 \Phi(r_0^2 R) - \frac{1}{2A} \int_s \Sigma^{ij} g_{ij} dA \right] = -\frac{\Delta G_{S0}}{\Delta T} (1 + r_0^2 R) \frac{g_0^{ij} \nabla_k g_{ij}}{(g_{mn} g_0^{mn})^2} - \frac{1}{\Delta T} \frac{1}{2A} \partial_k \int_s \Sigma^{ij} g_{ij} dA$$

Considering a 2-Sphere, the Ricci scalar is $R = 2/r_0^2$, since $\nabla_k R = 0$, $\nabla_k g_0^{ij} = 0$, and $k = r$

$$\left. \frac{\partial \Delta S_S}{\partial r} \right|^{sph} = \frac{\Delta G_{S0}}{\Delta T} \frac{-6 \left[\frac{r \sin^2 \theta}{r_0^2 \sin^2 \theta_0} + \frac{r}{r_0^2} \right]}{\left(1 + \frac{r^2}{r_0^2} + \frac{r^2 \sin^2 \theta}{r_0^2 \sin^2 \theta_0} \right)^2} - \frac{1}{\Delta T} \frac{r_0^2}{r^2} \Sigma_r(r)$$

where $\Delta G_{S0} = \gamma_0$, r_0 , Σ , and R are the surface energy in [$J.m^{-2}$], the radius at the thermodynamic state 0 (assumed as equilibrium or initial condition vector radius for Equilibrium, i.e., $\partial \Omega = g_0 \quad_{ij}$) in [m], the surface stress tensor in [$N.m^{-1}$], and the Ricci scalar, respectively.

Experimental Validation, $\Gamma = f(\nabla T)$: SDAS



Application Example 1

- **Experimental validation** [Ferreira et al. (2025)]:

SDAS can be calculated in terms of Gibbs-Thomson which is the Thermal Field Tensor Γ as,

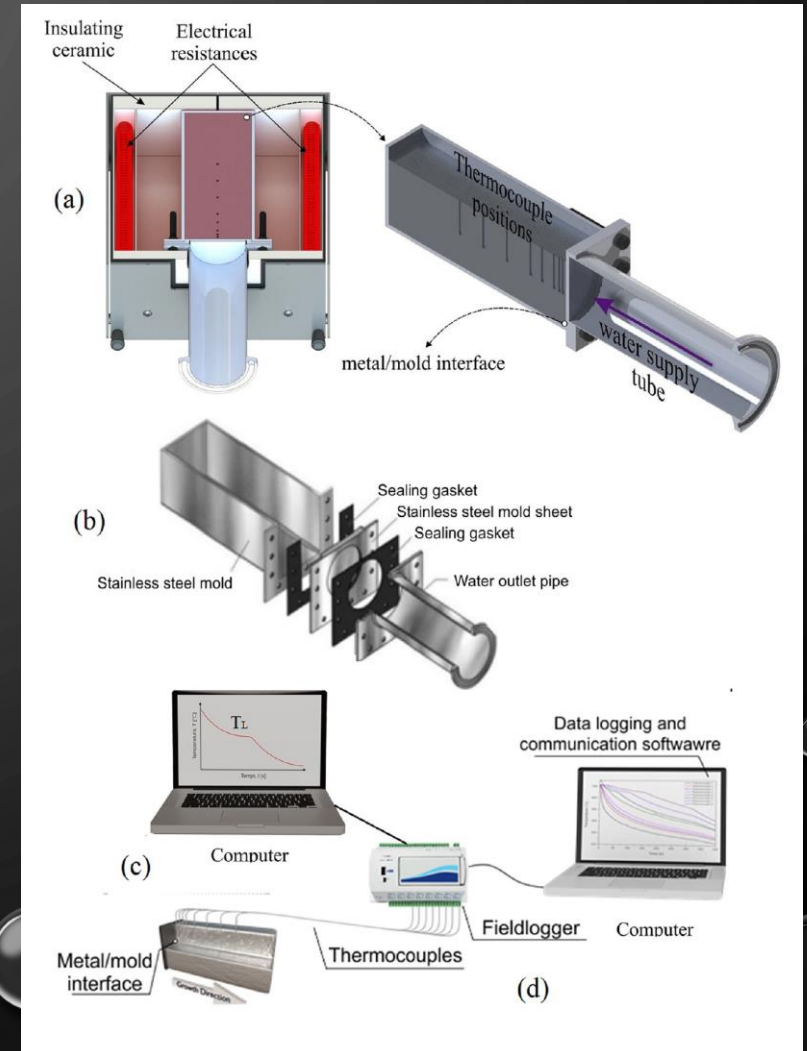
$$SDAS = 5.5 (M \cdot t_{SL})^{\frac{1}{3}}$$

$$M = \frac{-\Gamma \sum_{j=1}^n w_j \beta_j}{\sum_{j=1}^n m_j \left(1 - \frac{k_j}{k_j + (1 - k_j) \exp\left(\frac{-v \delta_j}{D_j}\right)} \right) (c_{f,j} - c_{0,j})/D_j} \ln \left[\frac{\sum_{j=1}^n m_j \left(1 - \frac{k_j}{k_j + (1 - k_j) \exp\left(\frac{-v \delta_j}{D_j}\right)} \right) c_{f,j}/D_j}{\sum_{j=1}^n m_j \left(1 - \frac{k_j}{k_j + (1 - k_j) \exp\left(\frac{-v \delta_j}{D_j}\right)} \right) c_{0,j}/D_j} \right]$$

Application Example 1

- **Experimental validation** [Mendes et al. (2023)]:

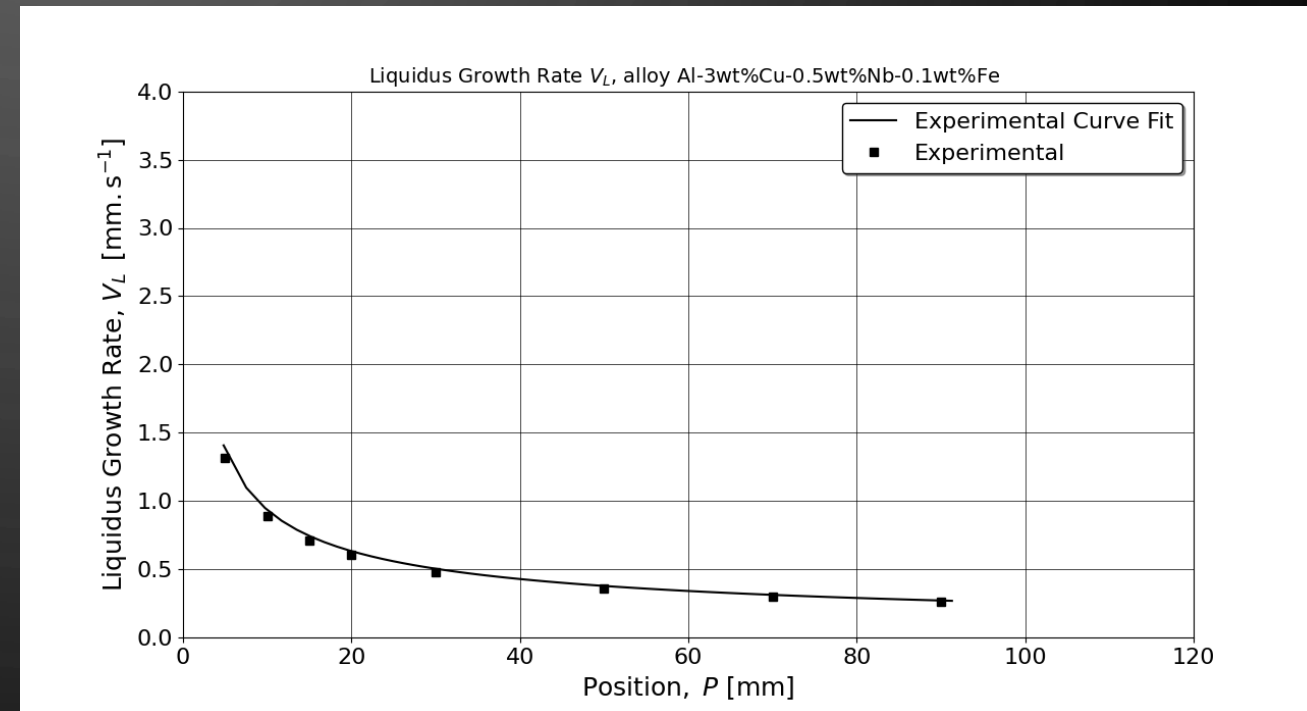
Figure 1 Schematic representation of transient horizontal solidification experiment.



Application Example 1

- **Experimental validation** [Mendes et al. (2023)]:

Figure 2 Experimental data from transient solidification measurements yielded **(A) dendrite tip growth rates**, (B) liquidus thermal gradients, and (C) dendrite tip cooling rates.

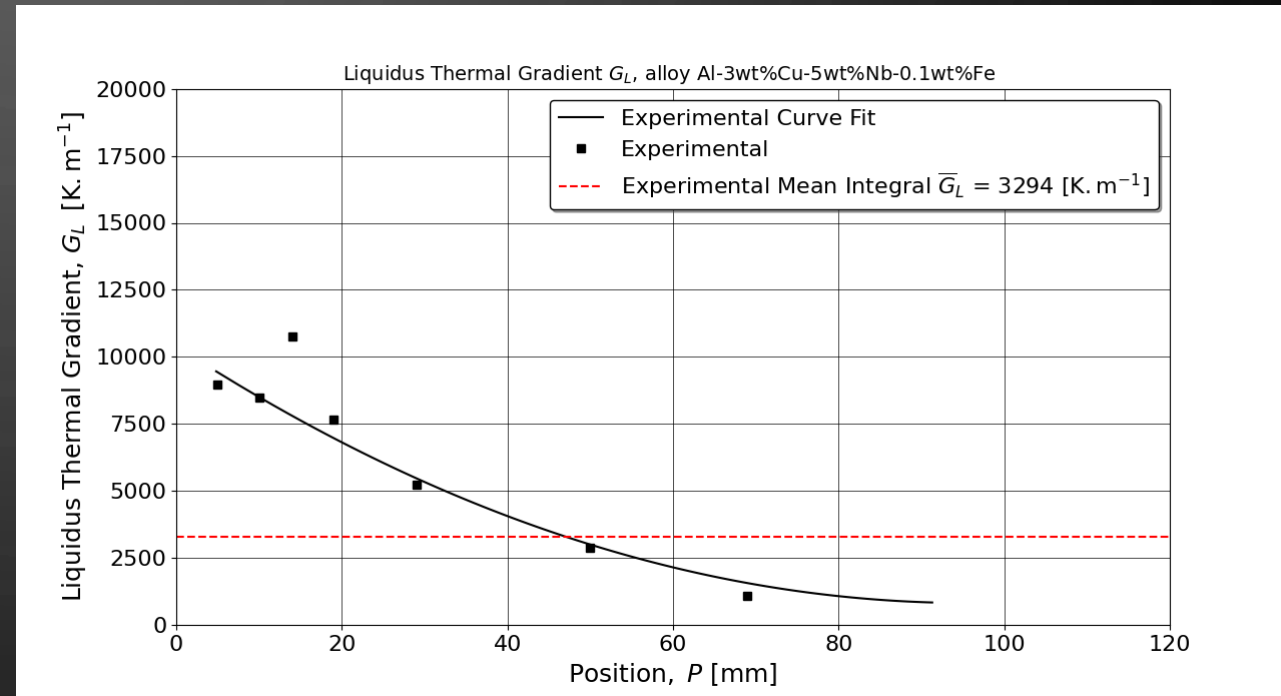


Application Example 1

- **Experimental validation** [Mendes et al. (2023)]:

Figure 3 Experimental data from transient solidification measurements yielded (A) dendrite tip growth rates, **(B) liquidus thermal gradients**, and (C) dendrite tip cooling rates.

$$\nabla T = 3294 \text{ K/m}$$

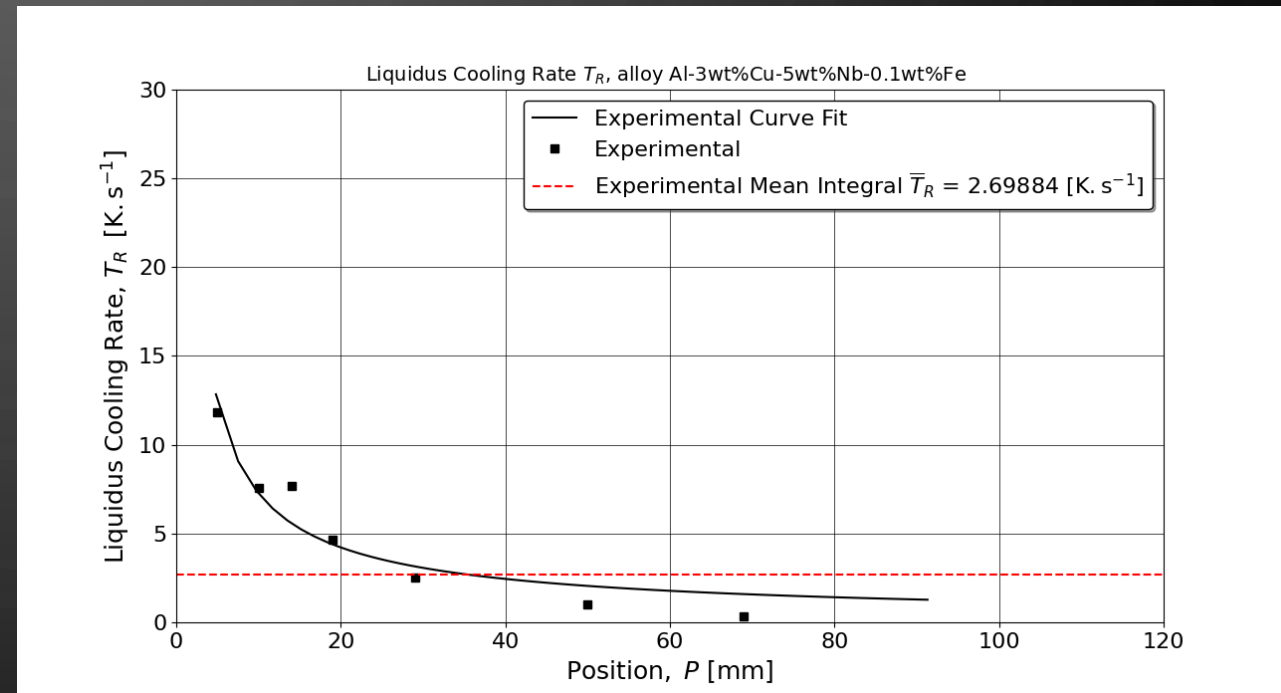


(B)

Application Example 1

- **Experimental validation** [Mendes et al. (2023)]:

Figure 4 Experimental data from transient solidification measurements yielded (A) dendrite tip growth rates, (B) liquidus thermal gradients, and **(C) dendrite tip cooling rates.**



(C)

Application Example 1

- Results of Nucleation Calculation [Present work]:

Table 1 – Calculated nucleation variables as a function of thermal gradient for FCC_A1 phase. alloy composition Al-3wt%Cu-5wt%Nb-0.01wt%Fe.

∇T $K.m^{-1}$	ΔT K	$r_{C.Hom}^{1st}$ m	$r_{C.Het}^{1st}$ m	θ^{1st} rad	$\frac{\partial f(\theta^{1st})}{\partial r}$ $1/m$	$r_{C.Hom}^{2nd}$ m	$r_{C.Het}^{2nd}$ m	θ^{2nd} rad	$\frac{\partial f(\theta^{2nd})}{\partial r}$ $1/m$	σ_{SL} $N.m^{-1}$	Σ $N.m^{-1}$	γ_{SL} $J.m^{-2}$
646.441	0.101843	6.26850E-06	9.86304E-07	0.38436	-0.06233	6.26850E-06	6.26848E-06	3.06101	-15.9999	-1.38990	-0.01380	0.31278
1012.45	0.153094	6.01650E-06	1.26170E-06	0.48007	-0.14755	6.01650E-06	6.01648E-06	3.05588	-15.9998	-1.50282	-0.12672	0.45128
2015.41	0.279223	5.51250E-06	1.65170E-06	0.63661	-0.43039	5.51250E-06	5.51248E-06	3.04991	-15.9998	-1.74361	-0.36751	0.75412
3001.66	0.389723	5.16600E-06	1.81812E-06	0.72559	-0.69747	5.16600E-06	5.16597E-06	3.04768	-15.9998	-1.92228	-0.54618	0.98639
3215.66	0.412416	5.10300E-06	1.84080E-06	0.74049	-0.75104	5.10300E-06	5.10297E-06	3.04738	-15.9998	-1.95605	-0.57995	1.03110
3294.00	0.420615	5.08089E-06	1.84819E-06	0.74563	-0.77012	5.08089E-06	5.08087E-06	3.04729	-15.9998	-1.96801	-0.59191	1.04701
3327.29	0.424099	5.07150E-06	1.85133E-06	0.74781	-0.77832	5.07150E-06	5.07147E-06	3.04725	-15.9998	-1.97309	-0.59699	1.05376
4068.77	0.499281	4.88250E-06	1.90362E-06	0.78993	-0.94828	4.88250E-06	4.88248E-06	3.04659	-15.9998	-2.07762	-0.70152	1.19434
5116.32	0.599473	4.66200E-06	1.94265E-06	0.83549	-1.15768	4.66200E-06	4.66198E-06	3.04610	-15.9998	-2.20481	-0.82871	1.36925
6012.88	0.680721	4.50450E-06	1.95725E-06	0.86587	-1.31256	4.50450E-06	4.50448E-06	3.04593	-15.9997	-2.29939	-0.92329	1.50230
7061.06	0.771435	4.34700E-06	1.96163E-06	0.89456	-1.47028	4.34700E-06	4.34698E-06	3.04587	-15.9997	-2.39735	-1.02125	1.64297
8029.62	0.851824	4.22100E-06	1.95823E-06	0.91635	-1.59758	4.22100E-06	4.22098E-06	3.04591	-15.9997	-2.47830	-1.10220	1.76159
9134.83	0.917265	4.12650E-06	1.95186E-06	0.93203	-1.69326	4.12650E-06	4.12648E-06	3.04599	-15.9998	-2.54062	-1.16452	1.85446
9435.16	0.940143	4.09500E-06	1.94904E-06	0.93714	-1.72513	4.09500E-06	4.09498E-06	3.04603	-15.9998	-2.56171	-1.18561	1.88620

Application Example 1

- Results of Nucleation Calculation [Present work]:

Table 1 – Calculated nucleation variables as a function of thermal gradient for FCC_A1 phase. alloy composition Al-3wt%Cu-5wt%Nb-0.01wt%Fe.

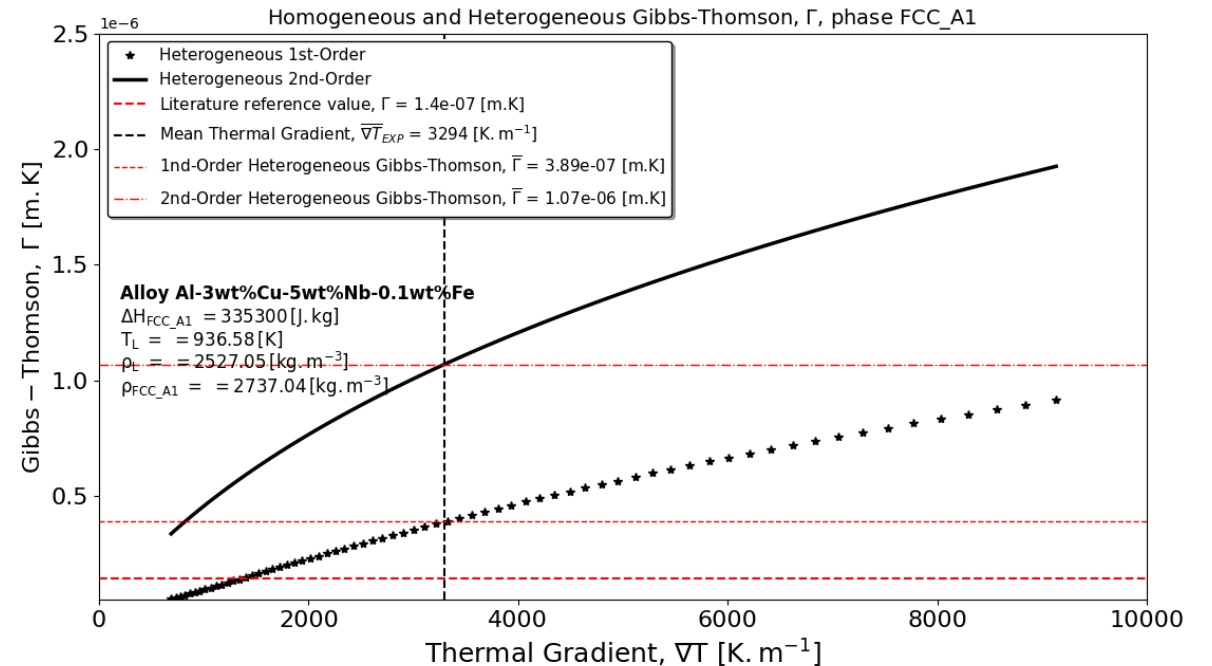
∇T $K.m^{-1}$	ΔT K	$\frac{\partial \gamma_{SL}}{\partial r}$ $J.m^{-3}$	ΔS_V $J.m^{-3}K^{-1}$	Γ^{1st} $m.K$	Γ^{2nd} $m.K$	$\frac{\partial \Delta S_V}{\partial r}$ $J.m^{-4}K^{-1}$	ΔS_S $J.m^{-2}K^{-1}$	ΔS_C $J.m^{-3}K^{-1}$	ΔG_V $J.m^{-3}$	ΔG_S $J.m^{-2}$	ΔG_C $J.m^{-3}$
646.441	0.101843	-534448	-1080580	5.02241E-08	3.19203E-07	-1080580	-5247750	-12.28470	-110050	0.312778	-1.25111
1012.45	0.153094	-565334	-1591090	9.65786E-08	4.60546E-07	-1591090	-3692720	-11.79080	-243586	0.451275	-1.80510
2015.41	0.279223	-639538	-1964060	2.30596E-07	7.69613E-07	-1964060	-2290420	-10.80310	-548412	0.754119	-3.01648
3001.66	0.389723	-703186	-2055790	3.54281E-07	1.00666E-06	-2055790	-1804320	-10.12400	-801188	0.986394	-3.94558
3215.66	0.412416	-716153	-2066180	3.79587E-07	1.05228E-06	-2092930	-1736480	-10.00060	-852126	1.0311	-4.12440
3294.00	0.420615	-720831	-2069460	3.88700E-07	1.06852E-06	-2100810	-1713940	-9.95726	-870458	1.04701	-4.18802
3327.29	0.424099	-722818	-2070860	3.92572E-07	1.07541E-06	-2100980	-1704360	-9.93885	-878249	1.05376	-4.21506
4068.77	0.499281	-765565	-2092930	4.75221E-07	1.21888E-06	-2103060	-1533330	-9.56846	-1044960	0.312778	-4.77735
5116.32	0.599473	-822246	-2108820	5.82285E-07	1.39738E-06	-2108820	-1371620	-9.13634	-1264180	0.451275	-5.47699
6012.88	0.680721	-868073	-2115640	6.66172E-07	1.53316E-06	-2115640	-1275230	-8.82768	-1440160	0.754119	-6.00918
7061.06	0.771435	-919179	-2119740	7.56636E-07	1.67672E-06	-2119740	-1191520	-8.51902	-1635240	0.986394	-6.57186
8029.62	0.851824	-964467	-2121520	8.34037E-07	1.79778E-06	-2121520	-1132240	-8.27209	-1807160	1.0311	-7.04636
9134.83	0.917265	-1001360	-2122150	8.95194E-07	1.89256E-06	-2122250	-1091680	-8.08689	-1946580	1.04701	-7.41782
9435.16	0.940143	-1014270	-2122250	9.16196E-07	1.92495E-06	-2122300	-1078840	-8.02516	-1995220	1.05376	-7.54480

Application Example 1

- **Experimental validation** [Present work]:

Figure 5 Thermal field tensor (Gibbs-Thomson) function of thermal gradient.

∇T $K.m^{-1}$	ΔT K	$\frac{\partial \gamma_{SL}}{\partial r}$ $J.m^{-3}$	ΔS_V $J.m^{-3}K^{-1}$	Γ^{1st} $m.K$	Γ^{2nd} $m.K$
646.441	0.101843	-534448	-1080580	5.02241E-08	3.19203E-07
1012.45	0.153094	-565334	-1591090	9.65786E-08	4.60546E-07
2015.41	0.279223	-639538	-1964060	2.30596E-07	7.69613E-07
3001.66	0.389723	-703186	-2055790	3.54281E-07	1.00666E-06
3215.66	0.412416	-716153	-2066180	3.79587E-07	1.05228E-06
3294.00	0.420615	-720831	-2069460	3.88700E-07	1.06852E-06



Application Example 1

- **Experimental validation** [Ferreira et al. (2025)]:

SDAS can be calculated in terms of Gibbs-Thomson which is the Thermal Field Tensor Γ as,

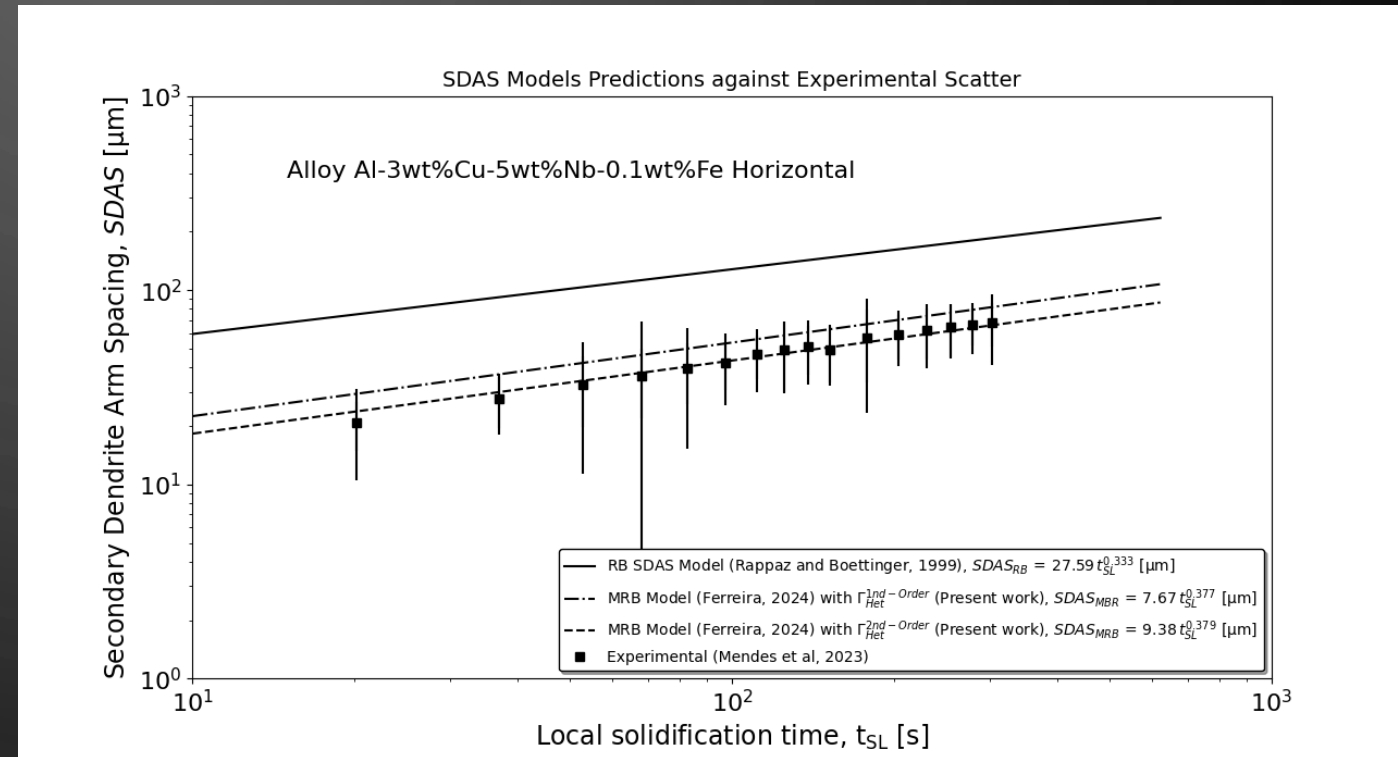
$$SDAS = 5.5 (M \cdot t_{SL})^{\frac{1}{3}}$$

$$M = \frac{-\Gamma \sum_{j=1}^n w_j \beta_j}{\sum_{j=1}^n m_j \left(1 - \frac{k_j}{k_j + (1 - k_j) \exp\left(\frac{-v \delta_j}{D_j}\right)} \right) (c_{f,j} - c_{0,j})/D_j} \ln \left[\frac{\sum_{j=1}^n m_j \left(1 - \frac{k_j}{k_j + (1 - k_j) \exp\left(\frac{-v \delta_j}{D_j}\right)} \right) c_{f,j}/D_j}{\sum_{j=1}^n m_j \left(1 - \frac{k_j}{k_j + (1 - k_j) \exp\left(\frac{-v \delta_j}{D_j}\right)} \right) c_{0,j}/D_j} \right]$$

Application Example 1

- **Experimental validation** [Present work]:

Figure 6 Comparison between secondary arm-spacing models and experimental scatter.



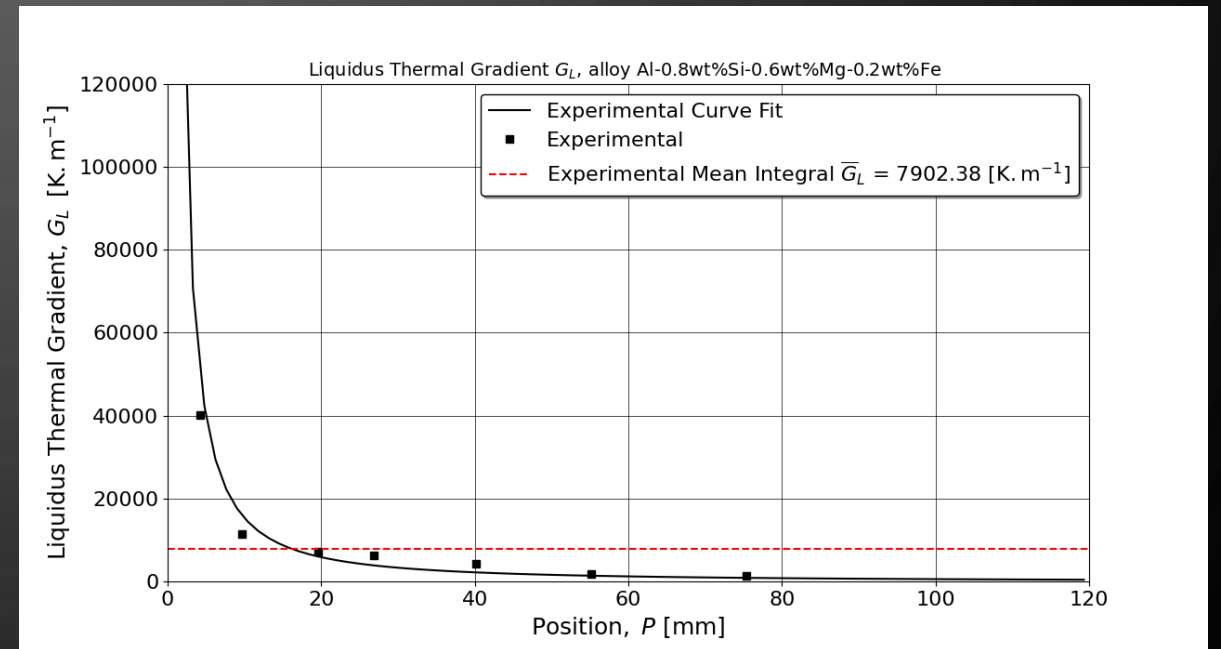
(A)

Application Example 2

- **Experimental validation** [Marques et al. (2025)]:

Figure 15 Experimental data from transient solidification measurements yielded (A) dendrite tip growth rates, (B) liquidus thermal gradients, and (C) dendrite tip cooling rates.

$$\nabla T = 7902.38 \text{ K/m}$$

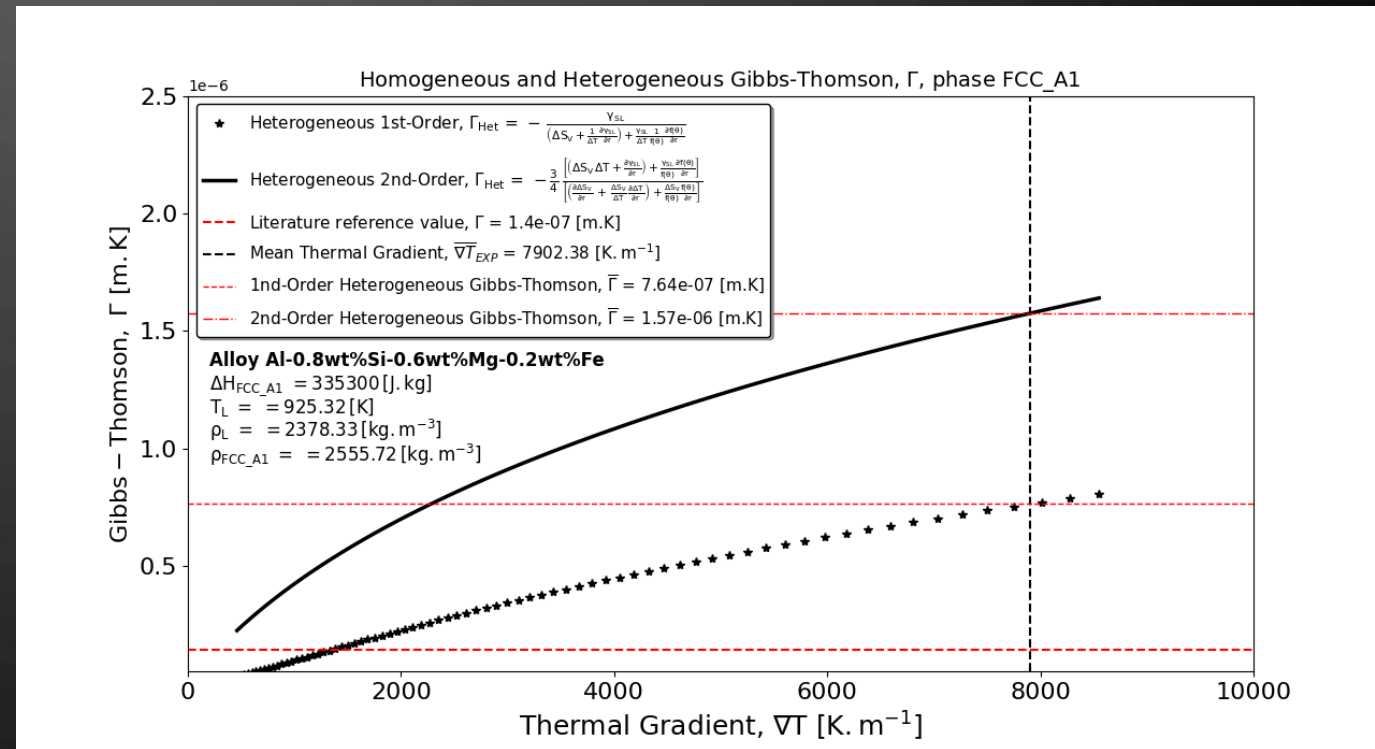


(B)

Application Example 2

- **Experimental validation** [Marques et al. (2025)]:

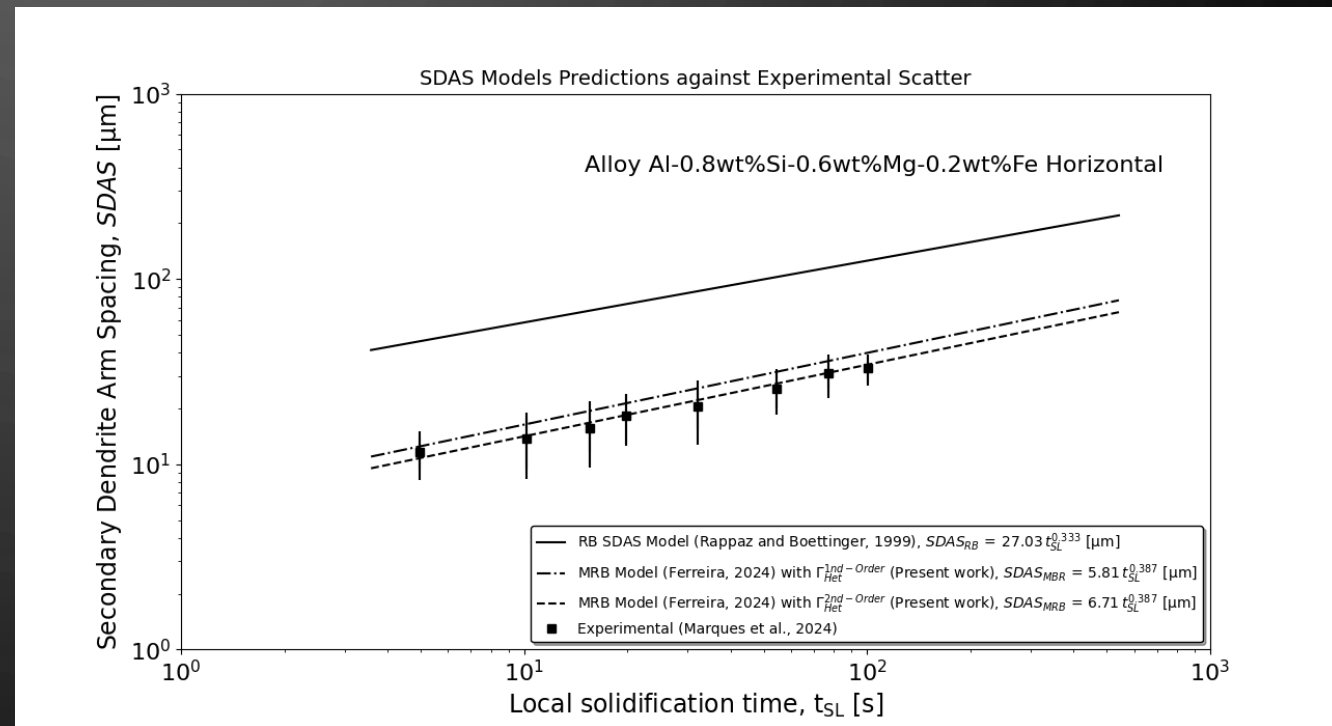
Figure 16 Thermal field tensor (Gibbs-Thomson) function of thermal gradient.



Application Example 2

- **Experimental validation** [Marques et al. (2025)]:

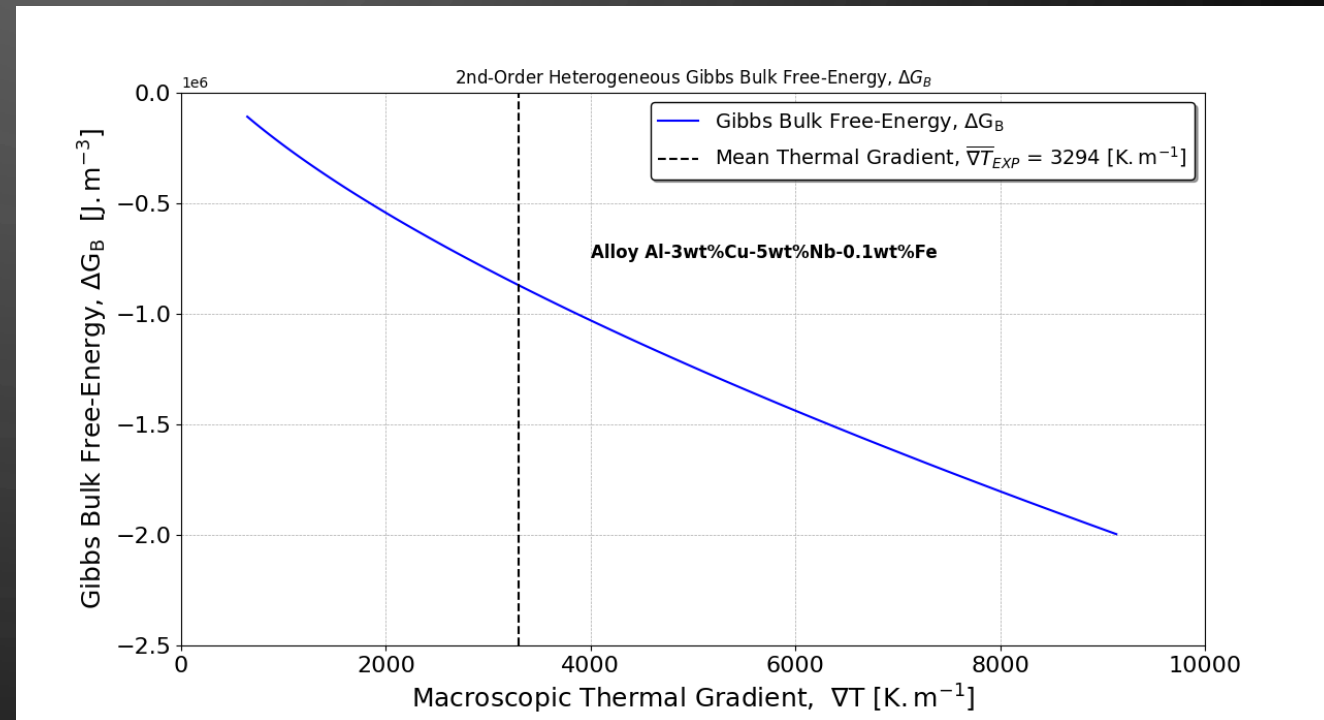
Figure 17 Comparison between secondary arm-spacing models and experimental scatter.



Application Example 1

- **Experimental validation** [Ferreira et al. (2025)]:

Figure 7 Gibbs bulk free-energy as a function of thermal gradient.

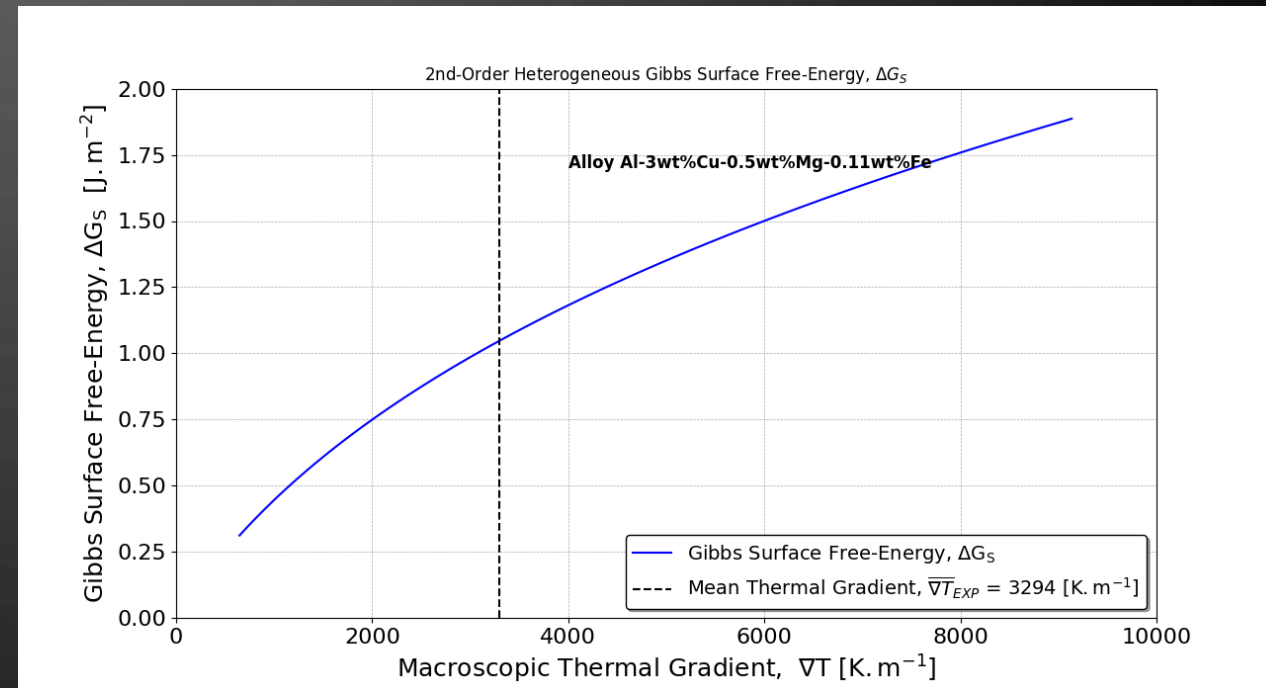


(A)

Application Example 1

- **Experimental validation** [Present work]:

Figure 8 Gibbs surface free-energy as a function of thermal gradient.

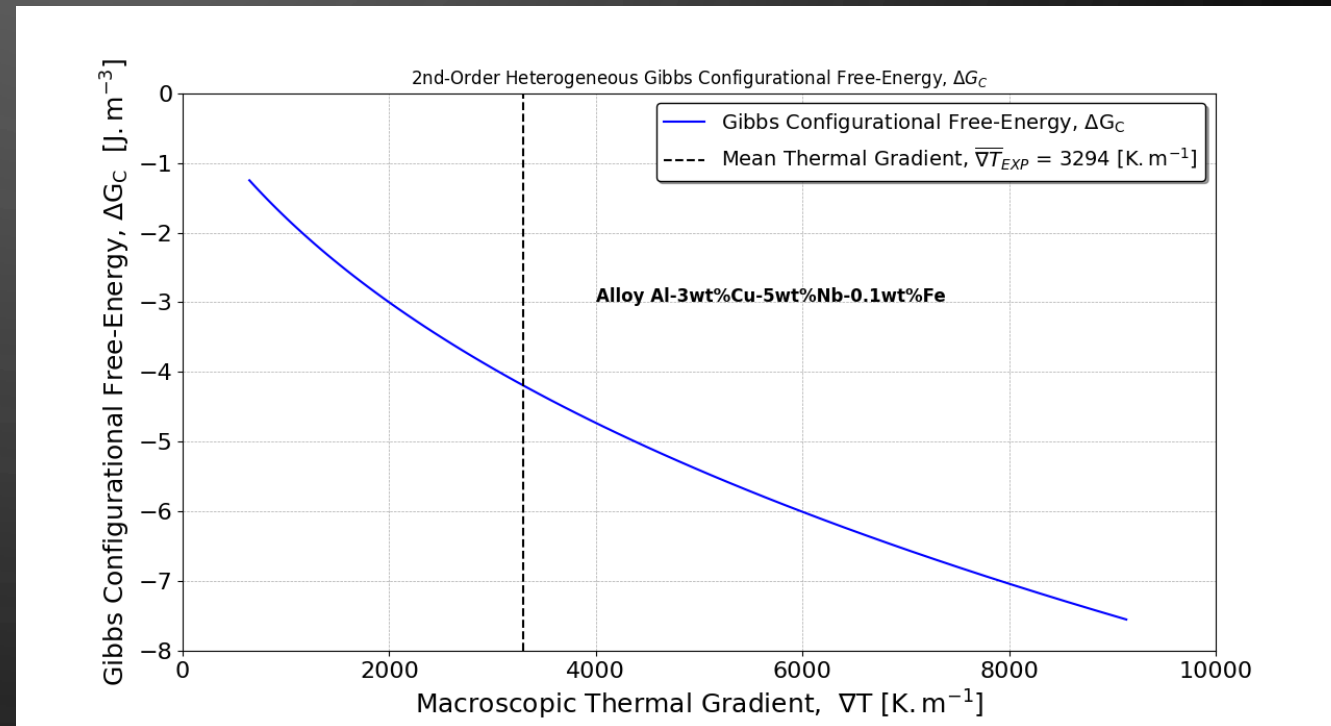


(A)

Application Example 1

- **Experimental validation** [Present Work]:

Figure 9 Gibbs configurational free-energy as a function of thermal gradient.

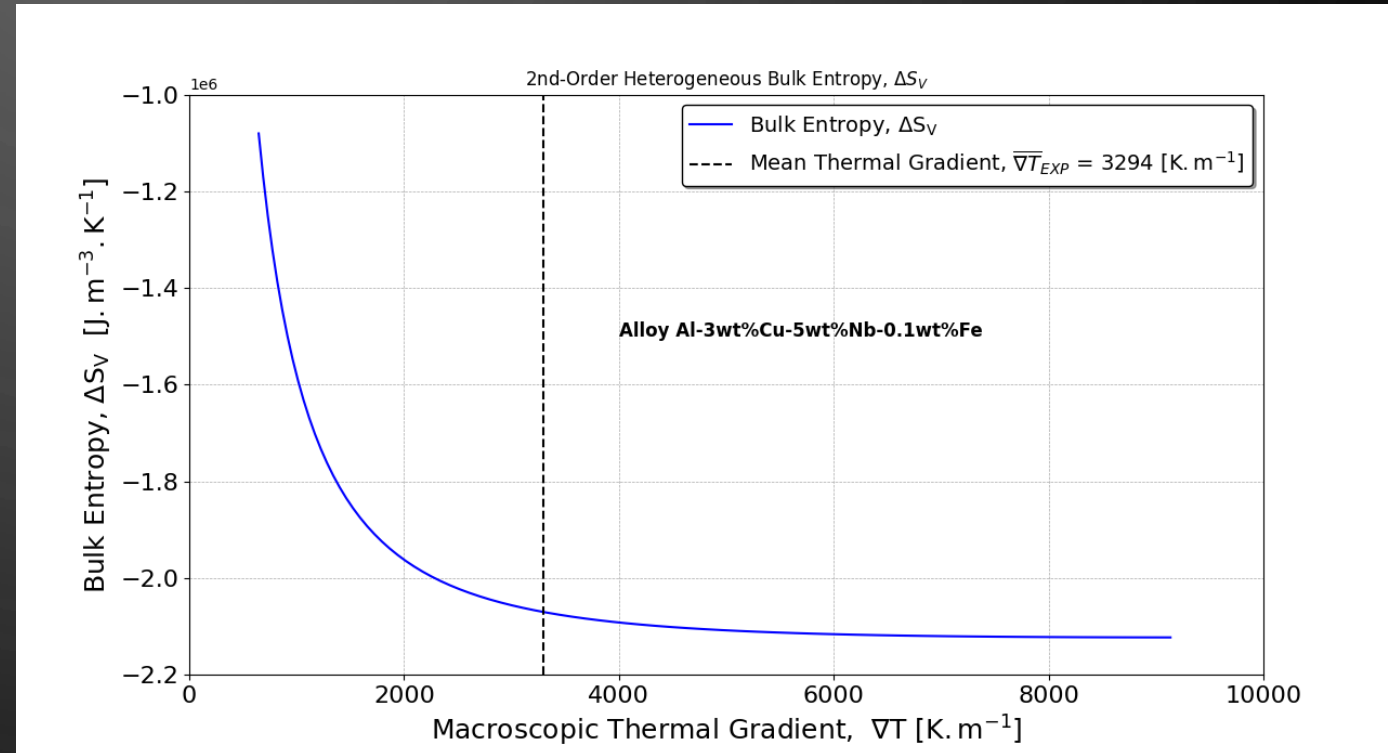


(A)

Application Example 1

- **Experimental validation** [Present work]:

Figure 10 Bulk entropy as a function of thermal gradient.

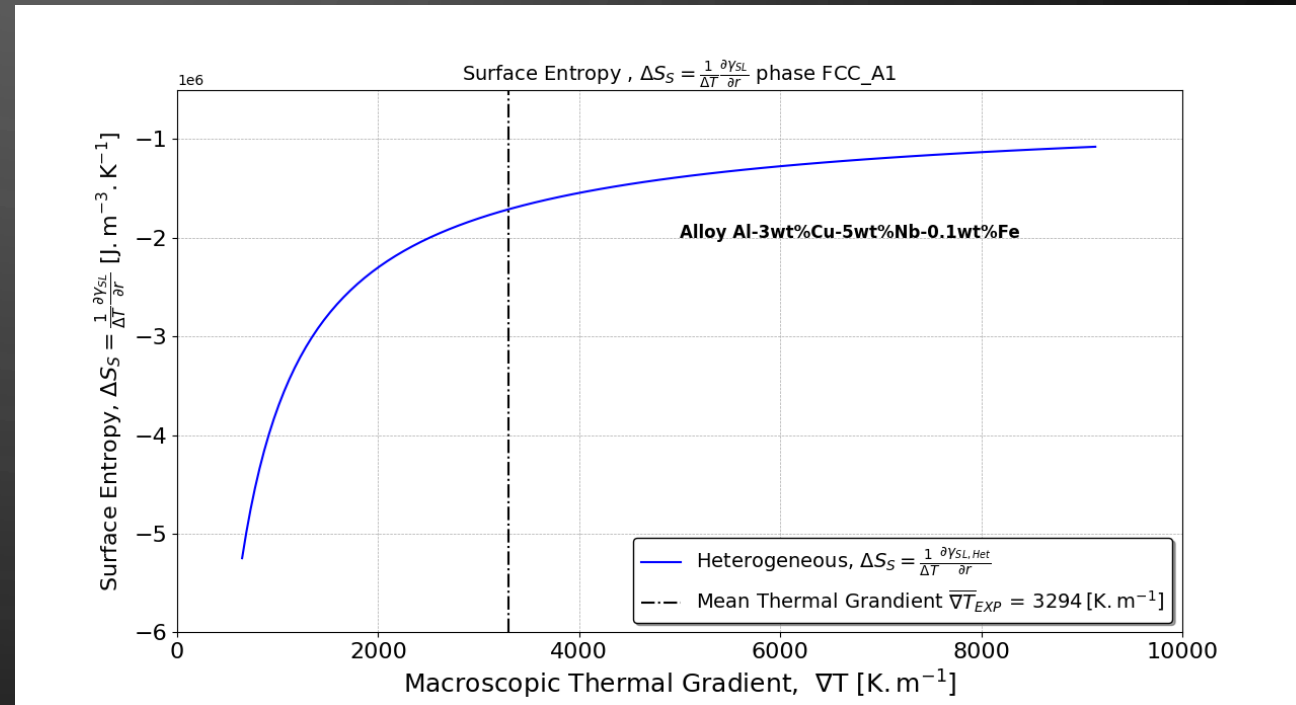


(A)

Application Example 1

- **Experimental validation** [Present work]:

Figure 11 Surface entropy as a function of thermal gradient.

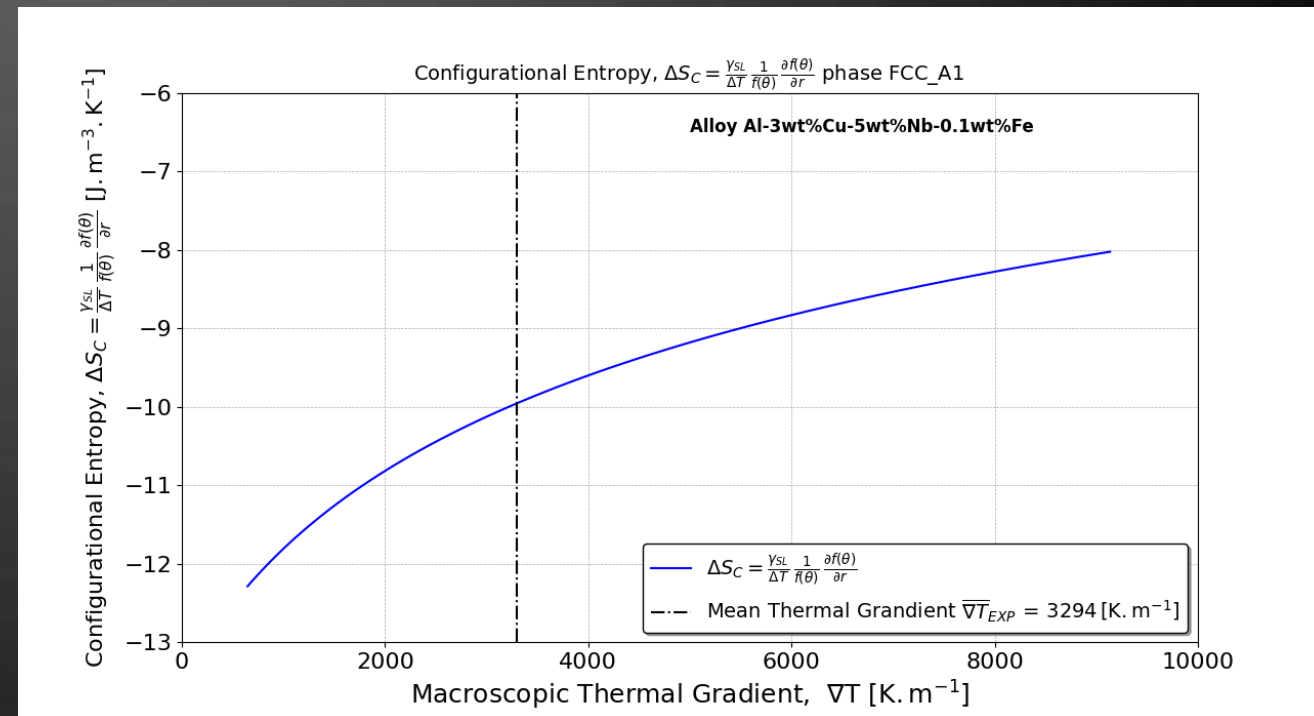


(A)

Application Example 1

- **Experimental validation** [Present work]:

Figure 12 Configurational entropy as a function of thermal gradient.



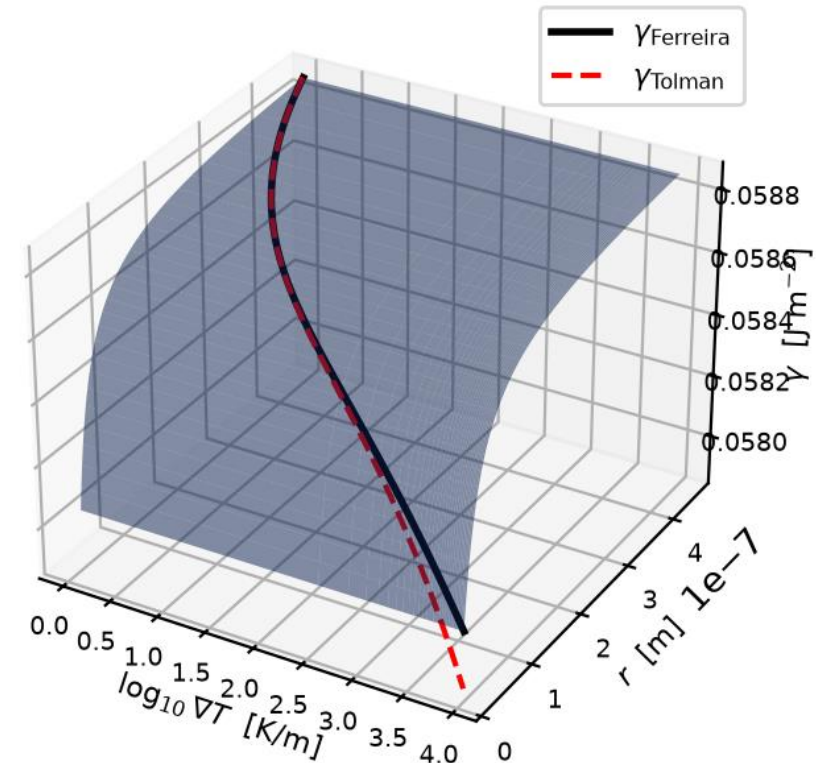
(A)

Application Example 3

- **Surface Energy Validation** [Present work]:

Figure 14 Vapour–liquid interfacial energy $\gamma(r, \nabla T)$ along the solved continuation. The full geometric surface-free-energy model $\gamma_{Ferreira}$ (black solid) is compared with the Tolman curvature law γ_{Tolman} (red dashed) over the solved domain (shaded). The two laws coincide in the planar, gradient-free limit γ_∞ and separate as r decreases and ∇T increases, the departure measuring the thermal-field coupling of the surface energy that the Tolman law omits. Axes: droplet radius r in m, decimal logarithm of the thermal gradient $\log_{10} \nabla T$ in $K m^{-1}$, and interfacial energy γ in $J m^{-2}$.

$\gamma(r, \nabla T)$ along the solved continuation



Conclusions

1. The second-order nucleation formulation, in relation to the critical radius, was validated and experimentally confirmed as a valid scientific theory;
2. The nucleation process analyzed has shown quadratic behavior in terms of nucleation radius;
3. Bulk and configurational free energies exhibit opposing signs relative to surface free energy;
4. Because the substrate's contribution aims to minimize local free energy, the configurational and bulk free energies share the same sign.

References

- [1] I.L. Ferreira Physica B: Condensed Matter. 695 (2024) 416494.
- [2] I.L. Ferreira: Materials. (2026) in press.
- [3] Y.-B. Kang, Calphad: Comput. Coupling Ph. Diagr. Thermochem. 50 (2015) 23
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Thanks for Watching!

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Dear Prof. Ivaldo Leão Ferreira,

On behalf of Ursula Von Der Leyen, President of the European Commission , I was asked to reply on your letter and of the attached presentation of your study aimed to present a suggestion to mitigate the production of greenhouse gases and stabilise climate change.

We examined the documentation attached and we took note of the content of your study, focused on complex global thermodynamic processes, plus of the consequent conclusions on necessary measures of climate mitigation aimed to achieve greenhouse gas reduction and to revert the current severe climatic trends.